Structured Amortizing Bonds

Fixed Income Amortizing-Stepup-Sinking-Bond — Generated: 2026-10-09 14:00 UTC

Inputs

Field	Value
Model	Structured Amortizing Bonds
Settlement / Value Date	2026-08-30
Dated Date / Accrual Start	2026-06-20
First Coupon Date	2026-12-20
Penultimate Coupon Date	2040-12-20
Maturity / Terminating Date	2041-06-20
Notional	1000000
Coupon Rate	0.0500
Pricing Basis	yield
Yield to Maturity	0.0600
Market Clean Price (% of Par)	100.00
Principal Schedule Type	straight_line
Cashflow Schedule	30 payment rows
Coupon Frequency (Payments / Year)	2
Day Count Convention	ACT/ACT ISMA
Scenario Shock (bp)	25

Results

Metric	Value
Fair Value (Clean)	$943,393.01
Accrued Interest	$9,699.45
Fair Value + Accrued	$953,092.46
Yield to Maturity	6.0000%
Duration	5.8385
Modified Duration	5.6684
BPV (+1bp Price Change)	$-540.25
Convexity	50.7291

Analysis

The latest run includes parallel-rate clean-value scenarios, principal runoff, and the contractual coupon profile by payment date.

Methodology

The engine generates deterministic coupon, principal and fixed-payment cashflows from the contractual schedule. Each cashflow is discounted using either the supplied yield to maturity or the interpolated zero curve. Accrued interest is separated from dirty value to produce clean value, and duration, modified duration, convexity and BPV are calculated consistently from the same cashflows and pricing basis.

Stage	Implementation
Schedule construction	Generate contractual payment dates and apply the coupon and principal terms in effect for each period.
Cashflow calculation	Calculate accrual, coupon, principal redemption and any additional fixed payment, then roll opening notional forward.
Valuation	Discount each cashflow using the selected yield or continuously compounded zero curve and separate accrued interest.
Risk diagnostics	Reprice under parallel shocks and calculate yield, duration, modified duration, convexity, BPV and principal runoff.

References: docs/methodology/amortizing_stepup_sinking_bond.md, Math/GenericBonds.html, Math/lcb.html, Math/BondTable.html, Math/Daycount.html, Math/Dategen.html

Testing Scope

Test Area	Coverage
Contract dates	Settlement, dated date, first and penultimate coupon dates, maturity and stub-period generation.
Cashflow schedules	Bullet, straight-line and custom sinking principal; level, step-up and step-down coupons; fixed payments.
Valuation	Yield-based and curve-based discounting, accrued interest, clean/dirty reconciliation and yield solving.
Risk outputs	Parallel-rate scenarios, duration, modified duration, convexity, BPV and principal runoff.
Independent benchmark	Published GenericBonds bullet and straight-line examples under the documented default terms.

Testing Results

Check	Observed	Acceptance Criterion	Status
Cashflow identity	Maximum residual 0.000e+00	Coupon + principal + fixed payment = total cashflow	Pass
Principal roll-forward	Maximum link error 0.000e+00; terminal residual 0.000e+00	Consecutive notionals reconcile and principal is fully repaid	Pass
Clean/dirty reconciliation	Residual 0.000e+00	Clean value + accrued interest = dirty value	Pass
Parallel-rate direction	Up-rate PV <= base PV <= down-rate PV	Positive-rate shock lowers value; negative-rate shock raises value	Pass

Internal checks are calculated from this saved pricing run. The external comparison is included only when the saved dates, notional, coupon, frequency, day count, yield and principal style match the documented benchmark.

External Benchmark Comparison

External GenericBonds benchmark - straight-line amortization

Metric	DerivaPro	External	Difference	Status
Fair Value (Clean)	$943,393.01	$943,393.01	+0.000292	Reconciled
Accrued Interest	$9,699.45	$9,699.45	+0.003552	Reconciled
Fair Value + Accrued	$953,092.46	$953,092.46	+0.003844	Reconciled
Duration	5.8385	5.8385	-0.000019	Reconciled
Modified Duration	5.6684	5.6684	+0.000029	Reconciled
BPV (+1bp Price Change)	$-540.25	$-540.25	-0.003649	Reconciled
Convexity	50.7291	50.7291	-0.000009	Reconciled

Limitations

This report reflects a single pricing run under the stated assumptions. Model outputs should be independently reviewed before use in production or commercial decision-making.